

COMBSS: Scalable Best Subset Selection for Generalised Linear Models

An interactive demonstration — StatFest 2026

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Abstract

Best subset selection — identifying the optimal k predictors from p candidates — is fundamental for building interpretable and parsimonious statistical models, but the underlying combinatorial problem is NP-hard.

COMBSS (Continuous Optimisation for Best Subset Selection) overcomes this barrier by reformulating the discrete problem as a continuous optimisation, making it scalable to high-dimensional settings where p far exceeds n .

This presentation walks through the framework for linear, logistic, and multinomial regression, illustrates its performance on simulated data and two real biomedical applications — cancer gene-expression classification ($p = 2,308$ genes) and a rice GWAS ($p \approx 158,000$ SNPs) — and demonstrates the open-source `combss` (R and Python) packages with practical examples.

Collaborators

Name	Affiliation
Sarat Moka (<i>presenter</i>)	UNSW Sydney
Zdravko Botev	UNSW Sydney
Benoit Liquet	Université de Pau et des Pays de l'Adour & Macquarie University
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Name	Affiliation
Houying Zhu	Macquarie University

How this demonstration is structured

This site is the talk. Each section below has its own page; navigate via the top menu or click through in order.

Section	Pages
Motivation — what sparse-constrained GLM is, and why lasso / MIO are not the last word	Sparse GLM → Lasso → MIO → COMBSS preview
Methodology — Boolean relaxation, Frank-Wolfe iterates, homotopy in the curvature parameter	Relaxation → Frank-Wolfe → Homotopy
Demos — five runnable demos in R and Python	Linear sim · HD logistic · Khan SRBCT · Rice GWAS · Comparisons
Use it yourself	Install · References

Packages

- **R** — [combss on CRAN](#) — `install.packages("combss")`
- **Python** — [combss on PyPI](#) — `pip install combss`

Source

All code and slides for this presentation are at <https://github.com/saratmoka/combss-statfest>.